

1. INTRODUCTION

This chapter reviews issues related to the analysis of psychophysiological data. It focuses on general questions, which are relevant to a variety of techniques, rather than specific issues related to individual methods. Another chapter in this book (Jennings & Stine, this volume) also deals with statistical issues in psychophysiology, and focuses on issues of inference and data interpretation.

1.1. Stages of data processing

In describing the procedures used in analyzing psychophysiological data, it is useful to distinguish among different stages of analysis. The **first stage** is *signal-enhancement* and elimination of observations that are artifactual or can be considered outliers. This stage, sometimes called “signal-conditioning,” involves, at least in part, techniques that are specific to each type of physiological measure.

The **second stage** involves *data reduction* (sometimes called “quantification” or “parameter extraction”). Most psychophysiological experiments include a large number of observations per subject (i.e., dependent variables). Whereas this provides data sets that are rich in information content, it also increases the probability of spurious or noisy observations. In general, one of the main objectives of signal processing is maximizing the signal-to-noise ratio. A large number of observations may contain many different signals to study, but may also contain a large amount of noise. It is advantageous, therefore, to select information that is relevant and to minimize redundancy between the dependent variables entered in the statistical analysis. Redundant dependent variables will not necessarily increase the signal, yet they may generate additional opportunities for noise to be present. An important step in signal processing is to reduce the data set to the smallest possible number of dependent variables (each variable affected by the smallest possible amount of noise) while preserving the information content of the original data set as much as possible. However, a moderate degree of redun-

dancy may be useful in distinguishing random noise from systematic noise, thus making it possible to derive a valid and reliable estimate of the “signal.”

The **third stage** of data processing is *statistical analysis*, which may include hypothesis testing, model fitting, parameter estimation, and so forth. As mentioned earlier most psychophysiological experiments are multivariate in nature – that is, they involve repeated observations from the same subjects. The multivariate nature of psychophysiological measures is one of their greatest assets: it allows for the experimenters to view phenomena from different “points of view,” thus affording a more complete and detailed picture of bodily phenomena. However, it often introduces special types of problems in the statistical evaluation of data sets. Two general approaches can be considered: the *univariate approach*, which considers one dependent variable at a time, and the *multivariate approach*, which considers all dependent variables, as well as their covariation, together (see Huberty & Morris, 1989). Each of these approaches has advantages and disadvantages. In general, when there is a limited number of subjects in the study (relative to the number of dependent variables), and the dependent variables tend to be correlated with each other across conditions (or groups) in the same manner as within conditions (or groups), the univariate approach is more powerful. The multivariate approach is preferable when there is a large number of subjects in the study (relative to the number of dependent variables), and the within- and between-condition (or group) correlations among dependent variables add important information, as it is the when the dependent variables tend to go in opposite directions. (The statistical analysis stage is treated in the chapter by Jennings and Stine, and will not be considered in detail here).

In concluding this section, it is important to remind the reader that the output of an analysis is only as good as its input (this is captured by the well-known aphorism “garbage in, garbage out”). Complex and sophisticated data analyses in no way can be substituted for clear experimental hypotheses, elegant experimental designs, or accurate data collection procedures. However, they can

complement these steps by providing clear and convincing data.

1.2. General issues

Psychophysiological data vary along a number of dimensions, and different types of data propose different types of analytical problems and solutions to the investigator. However, there are a few overarching themes that are valid for most types of measurements. Some general issues are particularly important and worth mentioning early in this chapter. **First**, *psychophysiological measures are typically indirect*. In other words, they are measures of events occurring outside the human body, which are related to events that occur inside the body. For instance, measures of the electrical activity produced by the body (such as the electrocardiogram (EKG) or the electroencephalogram (EEG)) are in fact measures of the difference in potential between two electrodes located on the surface of the body. Another example is positron emission tomography (PET), which measures the arrival of high-energy (gamma) photons to particular detectors located at some distance from the body, which, in turn, is related to the concentration of radioactive substances in various areas of the body. In general, psychophysiological measures are subject to some transformation with respect to the original signal generated within the body because of physical, physiological, or psychological reasons. This transformation may partially distort the signal (as well as its statistical properties) and may occasionally generate artifacts that need to be recognized and eliminated from the data before the statistical analysis is performed.

Second, most of the *data are sampled in a discrete fashion* both temporally and spatially. The transformation of analog parameters into discrete measures may distort the signal and require specific solutions to statistical problems. One of the major problems is obtaining a representative sample of the variable of interest, both over time and space. In addition, some techniques (e.g., measures of heart rate acceleration or deceleration) may be based on the observation of internal events (e.g., the occurrence of an R-wave in the EKG) whose spacing is variable (and, in the case of heart rate changes, is a function of the variable of interest). This may make it more difficult to compare observations across trials and subjects. Investigators using these measures have developed special approaches to address this problem (see the chapter on autonomic measures).

Third, most *psychophysiological measures are “multiply determined.”* In other words, measures are determined by the interaction of a variety of factors. Some of these factors may be extraneous to, and uncorrelated with, the experimental manipulations and, therefore, will contribute to the statistical noise. On the other hand, other factors may be related to the experimental manipulations in ways that are different from those intended by the researcher, which will contribute to both the statistical noise and the systematic effects. The latter type of factors are particularly insidious

and difficult to eliminate. One of the most important characteristics of good experimental methodology is the use of appropriate control procedures that distinguish between signal and systematic noise.

Fourth, most *psychophysiological measures are inherently noisy*, which make it difficult to distinguish between signal and noise in raw (single trial) observations. Several techniques have been developed to extract the signal from the noise. One of the simplest is signal averaging: a measure is repeated several times and the individual responses are then averaged together. Signal averaging is based on the central limit theorem, and can, therefore, be expected to provide reasonable estimates of the expected central tendency of the population in a large number of situations, provided that enough trials are used. Note that for signal averaging to work appropriately, it is important that the expected value of the noise be equal to zero across trials. Although this is a reasonable assumption in many cases, there are certainly situations in which it is not true. For instance, as we will see in section 3.2.2, frequency analysis (such as the Fourier transform) can be used to estimate the “power,” or variance, associated with a particular frequency. Because this value cannot be less than zero, the average of the power values for a particular frequency obtained in different trials (or in different subjects) will always be greater than zero, even if no systematic activity exists at that frequency.

A problem with signal averaging is that it eliminates potentially relevant information about the trial-to-trial (and in some cases subject-to-subject) variability of the signal. This variability is entirely attributed to stochastic phenomena. However, in most cases the variability of the signal is not entirely random, but depends on the influence of intervening variables. For instance, in an experiment that evaluates the size of the visual evoked potential as a function of stimulus intensity, the size of the response may also be influenced by subjects’ variables (e.g., allocation of attention or emotional factors), contextual conditions (e.g., stimulus sequence or noise in the environment), or other factors. When signal averaging is used, all of these intervening variables are treated as contributors to stochastic noise and are ignored. This may lead to loss of information or to systematic errors when contingencies exist between the experimental manipulations of interest and any of the intervening variables. Similar problems have been described with respect to other types of measures (see Estes, 1956; Siegler, 1987).

Another problem with signal averaging is that the signal may vary along dimensions other than intensity. In some cases, the latency of the phenomenon or its spatial location may vary from trial to trial (partly because of the reasons outlined above). In these cases signal averaging may produce systematic distortions, the most typical of which is a “smearing” of the response across different data points (or spatial locations). The effect of smearing is more intense for signals with a steep gradient. In cases of electrophysiological signals that vary quickly over

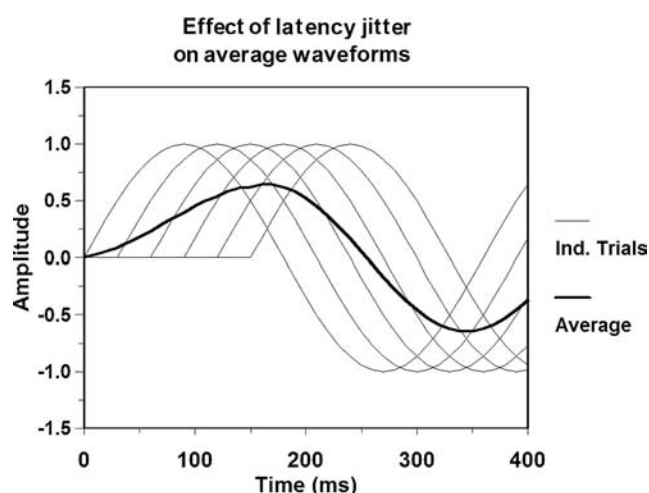


Figure 35.1. The effect of trial-to-trial variations in latency (latency jitter) on the amplitude of an averaged psychophysiological activity. In this example, the latency of the first positive peak varies systematically in latency between 90 and 270 ms. Although the amplitude of the peak on single trials (indicated by thin lines) is always equal to one, the amplitude of the peak of the average waveform (indicated by the thick line) is considerably smaller (slightly larger than 0.5). The duration of the activity also appears prolonged.

time, smearing leads to the phenomenon of “latency jitter”: the size of an evoked electrophysiological activity may be underestimated when using signal averaging because the activity “peaks” at different latencies on individual trials. A similar problem exists when data from different subjects are pooled together (for a discussion of this problem and some solutions, see Möcks, Köhler, Gasser, & Pham, 1988). Interestingly, the development of brain imaging methods, which have strong gradients in the spatial domain, has led to the recognition of a similar smearing problem in which average spatial maps are computed across subjects because of intersubject variability in brain anatomy. The effect of latency jitter is illustrated in Figure 35.1. Note that this problem is not specific to psychophysiological measures. For instance, investigators in psychology have long recognized that apparently smooth, speed-accuracy functions may result from the mixture of distributions of fast responses with a low level of accuracy and slow responses with a high level of accuracy in which the two distributions have some degree of partial overlap in response time (Yantis, Meyer, & Smith, 1991).

In order to address the smearing problem, several investigators have proposed to measure psychophysiological phenomena on single trials (or subjects). A critical step of this procedure is the development of techniques for identifying target features on single trial records. These techniques, although quite varied, can be grouped under the label of “pattern-recognition” algorithms. Some of them are based on visual inspection of individual data, and are most useful when the target features are much larger than the noise. Other techniques are “computer-controlled,” and based on statistical approaches such as

cross-correlation, auto-correlation methods, as well as other automatic methods. The individual records may then be aligned along the target feature and averaged together. Alternatively, some measure of the single record distribution other than the mean can be used. An example of an alignment methodology in the time domain is the Woody filter (Woody, 1967), which is a general-purpose statistical method; an example in the spatial domain is the alignment method proposed by Talairach and Tournoux (1988), a special purpose approach, which has become standard practice in brain imaging.

2. GENERAL CLASSIFICATIONS

Psychophysiological measures are usually defined as non-invasive measures of bodily functions taken with the intent of addressing issues that are relevant to psychologists and cognitive neuroscientists. This broad definition encompasses a large variety of measures that differ along a number of dimensions. Of these dimensions, some are important for the purposes of the present chapter because they result in different types of signal processing problems. I will pay particular attention to three of these dimensions.

2.1. Fast versus slow measures

As mentioned above, psychophysiological measures reflect bodily changes that are related to psychological phenomena. These changes occur over time. Some of them evolve very quickly (within a few milliseconds) and others quite slowly (over seconds, minutes, or even longer times). Indeed, it is possible to consider a parameter that will indicate the speed at which the activity evolves for each type of psychophysiological measure used. We can refer to this parameter as the “time constant” of a particular physiological measure. This time constant depends on three factors: (a) the rate of change of the relevant psychological phenomenon; (b) the time required by the physiological event being measured to track changes in the underlying psychological phenomenon; and (c) the time required for the measurement.

By and large, it is possible to consider individual time-delay functions for each of these steps, although in some cases interaction terms should also be considered. Note that each step may itself comprise several substeps. Each step can be considered as having an input, which comes from the previous step, and an output, which influences the next step. However, a delay will usually occur in the operation of translating an input into an output. An example of a function expressing this delay can be the following:¹

$$O_t - O_{t-1} = (I_t - O_{t-1}) * (1 - e^{-1/tc}) \quad (35.1)$$

¹ Note that this formula may not apply to all cases, and is only provided as an example of the typical influence of delaying factors on the time course of physiological measures. In fact, not only the time course but also the intensity of the output with respect to the input may vary from case to case.

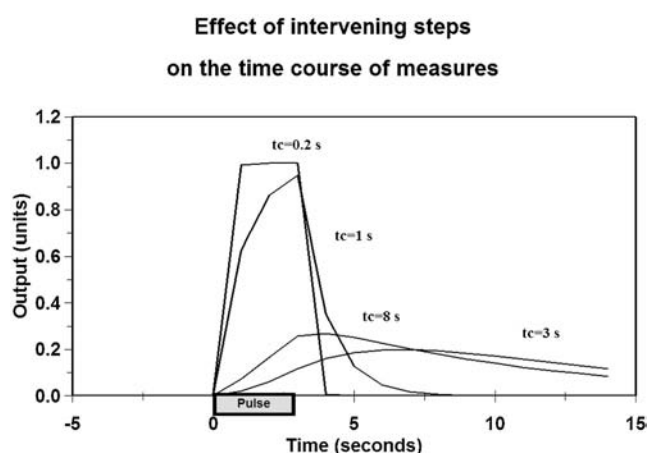


Figure 35.2. Simulated example of a series of individual steps between an input (pulse) and a final output measure. The input is a 3 second pulse (this may simulate a short duration stimulation). Each step introduces a delay, which is described by a parameter called time constant (see text). Four steps were simulated, with time constants, respectively, of 0.2, 1, 8, and 3 seconds.

In this function, the change in the output ($O_t - O_{t-1}$) of a particular step is a delayed function of the difference between its input I_t and its previous level O_{t-1} . The delay is determined by an exponential function whose critical parameter is tc , which is the *time constant* of that particular step. When several steps are involved, the total time-delay function is determined by a convolution of the individual functions. That is, the total time constant of a measurement is related to the integration of the time constants of each individual step. An example of this cascade, and of its effect on the final output measure, is illustrated in Figure 35.2. In this example, the initial input is a 3 second pulse (which is intended to simulate a brief stimulation). Four steps were simulated, with time constants of 0.2, 1, 8, and 3 seconds, respectively (for each step, the output serves as the input for the next step). Note that the final output is substantially delayed with respect to the input, and actually peaks well after the termination of the pulse (i.e., of the stimulation).

The steps contributing most to the total time constant are called “rate limiting” steps (step 3 and, to a smaller extent, step 4 in the example). Although all factors determining the measurement time constant (i.e., rate of change of the psychological event, delay of the physiological phenomena, and measurement time) are always present to some extent, there is often one (or a few) “rate limiting” factor. The nature of the rate limiting factor and the measurement time constant vary between different physiological measures. I will label measures with a long time constant “slow” measures and measures with a short time constant “fast” measures.

The following can be considered examples of slow measures: (a) measures of the electrodermal activity (EDA) in which the psychological or the physiological phenomena evolve slowly over time; (b) functional magnetic resonance imaging (fMRI) in which the physiological event

under study (e.g., a change in blood oxygenation) evolves slowly; and (c) PET (in which, in addition to a delay in the hemodynamic or metabolic response with respect to the triggering event, the actual measurement time is also relatively slow. Fast measures include (a) event-related brain potentials (ERPs); the magnetoencephalogram (MEG); (c) the electromyogram (EMG); and (d) the event-related optical signal (EROS). In all these cases, all three factors mentioned above are fast measures. Some other measures, for instance, heart rate and pupil diameter, are intermediate between these two extremes.

Measures with a short time constant have a good temporal resolution (i.e., the effects of two different psychological or physiological events occurring in rapid succession can be distinguished), whereas measures with a long time constant have a poor temporal resolution (i.e., events in rapid succession cannot be separated easily). This is illustrated in Figure 35.3 in which the time courses of the effect of two stimuli presented in rapid succession on measures with short and long time constants are compared.

Measures with a slow time constant provide few independent observations per unit time. In contrast, measures with a fast time constant provide a large number of independent observations over the same time unit. The processing of psychophysiological data usually requires transformation of the data into a series of numbers sampled over time (i.e., a time series). Then an issue arises: what is the optimal temporal sampling strategy for each type of measure? This sampling strategy must preserve the information contained in the signal as much as possible.

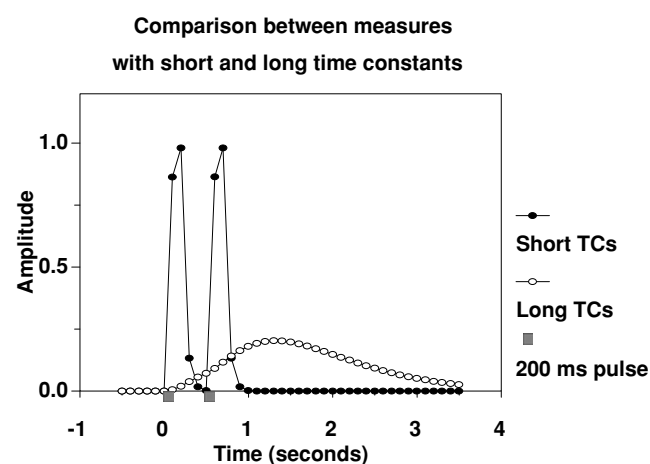


Figure 35.3. Simulated example of the response of output measures with short and long time constant to two stimulations in rapid succession. Two 200 ms stimulation pulses were presented, separated by an interval of 300 ms. This is intended to simulate two independent psychological events. The closed-circle line is obtained by assuming a cascade of three intervening steps between the stimulation and the measurement system, each introducing a delay characterized by a time constant of 200 ms. The open circle line is obtained by assuming a cascade of three intervening steps with time constants of 5 seconds. Note that short time-constant measure affords easy separation of the two pulses (i.e., psychological events), whereas the effects of the two pulses on the long time-constant measures are lumped together.

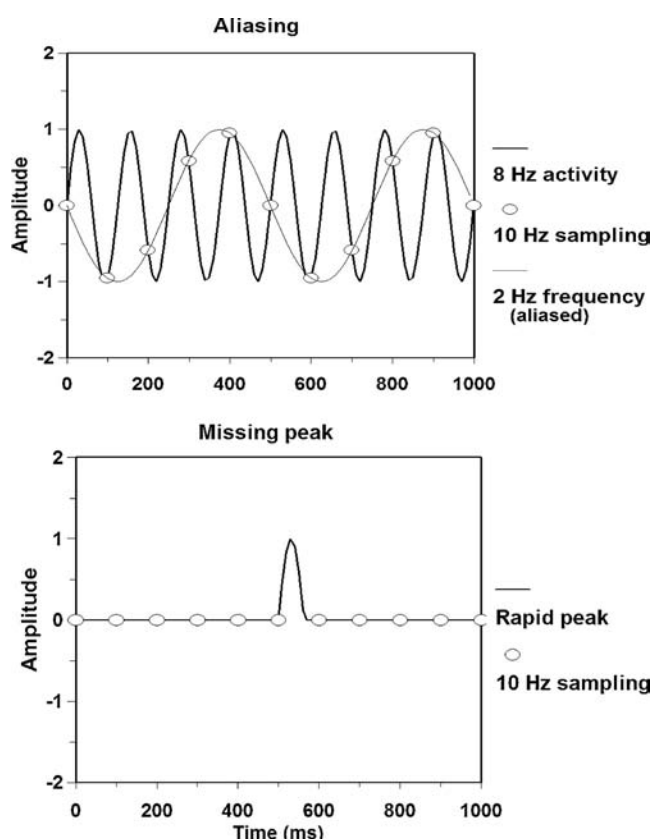


Figure 35.4. *Top:* The effect of sampling at a frequency lower than the Nyquist frequency (aliasing). The signal (indicated by the thick line) is sampled at 10 Hz (indicated by the open circles), which is less than the Nyquist frequency of 16 Hz. The result is an apparent 2 Hz signal (indicated by the thin line). *Bottom:* A rapid activity (duration = 62.5 ms) is measured using a 10-Hz sampling rate. In this particular case, the activity occurs in between two sampling points (open circles) and the peak is not detected by the measurement system.

The minimum sampling frequency at which information is maintained is twice the frequency of the signal, which is called the “Nyquist frequency.” At sampling rates slower than the Nyquist frequency, aliasing causes a high frequency signal to manifest itself as a lower frequency signal. This phenomenon is illustrated in the top portion of Figure 35.4. Also, a sampling frequency that is too slow may completely miss the occurrence of a very fast, transient phenomenon. This is illustrated in the lower portion of Figure 35.4. To avoid the risk of aliasing, investigators may wish to sample at the highest possible frequency. However, it should be remembered that, because each observation will contain both signal and noise, it is important to minimize the number of data points (i.e., dependent variables) entered into the statistical analysis. This may be achieved conveniently at the data reduction stage. Finally, some recording systems impose limitations on the maximum sampling rate. However, the issue of aliasing should always be considered when studying physiological variables. This problem is likely to occur when fast phenomena are studied using a relatively slow sampling rate (e.g.,

when EEG is measured at <20 Hz, EKG at <2 Hz, and respiration at <0.5 Hz).

2.2. Direct versus indirect measures

Psychophysiological measures also differ in the number of intervening steps that occur between the physiological phenomenon, which is the “real” target of the study (e.g., neuronal activity, sympathetic, or parasympathetic activation), and the actual physiological measure that is taken (e.g., scalp electrical activity, increased blood flow, increased heart rate, or increased EDA). The “indirect” nature of these measures may have several consequences. **First**, as we mentioned earlier, it may introduce a lag between the psychological event of interest and the observed changes. **Second**, each of the intervening steps may introduce distortions in the size or shape of the observations compared with the original signal.

A particularly important issue related to the indirect nature of psychophysiological measures is that of the “linearity” of the relationship between the measure and the underlying phenomenon of interest. A measure M is linearly related to an inferred psychophysiological variable θ if the following formula is valid:

$$M = k\theta + \text{offset} \quad \text{with } k \neq 0 \quad (35.2)$$

For all practical purposes, it may be sufficient that this formula is valid (at least approximately) for an interval of values of M and θ , provided that this interval contains the typically observed values of the measure and the expected range of the psychophysiological factor. Note that in order for the linearity assumption to be valid, two terms must remain constant across conditions in this formula: the proportionality factor k and the measurement *offset*. The proportionality factor k depends on variables such as the relationship between the psychophysiological phenomenon of interest and the physical phenomenon actually measured, which includes the propagation of the physiological measure to the surface, the recording and analysis procedures employed, and so forth. An example of a potential change in the proportionality factor is a change in the way neuronal signals are transformed into a hemodynamic phenomenon in young and old adults, which depends on the health of the subject’s cardiovascular system. Similarly, increases in neuronal activity at occipital and frontal locations may result in different increases in the scalp electrical signal measured over these areas because of different thickness of the skull in frontal and occipital regions. The offset factor may also depend on the effect of variables other than the physiological phenomenon of interest. For instance, an apparent increase in the amplitude of the ERP at frontal electrodes may be the consequence of an increase in the ocular artifact. Similarly, head movements may result in apparent large changes in the fMRI signal.

Another interesting example for psychophysiolgists is the difference between measures of EDA. Two such

measures are possible: (a) measures of the electrical *conductivity* between two locations on the skin, and (b) measures of the electrical *resistance* between the same locations. These two measures are inversely related to each other. However, the physiological phenomenon of interest (e.g., increase in conductivity associated with the activity of the sweat glands) is directly proportional to conductivity and inversely proportional to resistance. Therefore, in this example, conductivity is a linear measure of the physiological phenomenon, whereas resistance is not. Quigley and Berntson (1996) have pointed out that a similar case can be made for measures of heart rate compared to heart period, which are reciprocal to each other. Heart period, but not heart rate, appears to vary linearly with basic autonomic activation.

Linearity is useful because it allows for linear transformations of measure M (such as additions, subtractions, multiplications by a constant, and averaging), as well as the physiological variable θ , to be considered valid. Linear manipulations of the observed measures are commonly performed in the analysis of psychophysiological data. However, if the relationship between M and θ is nonlinear (i.e., if k or the offset are not constant), these types of operations, which are performed on the dependent variable, may not be valid with respect to the psychophysiological variable. Therefore, inferences should not be drawn because they may be misleading.

Lack of linearity may occur whenever any of the intervening steps between the observed measure and the psychophysiological construct of interest involve some form of nonlinear transformations. For example, consider the steps occurring between the onset of neuronal activity in response to a stimulus and the subsequent increased O^{15}_2 -PET response. These steps may include the release of some chemical by the active neurons, subsequent local vasodilation, increased regional blood flow, and, if a radioactive tracer is introduced in the blood flow, increased radioactivity in a particular volume of the head. In principle, the relationships between causes and effects that exist at each step may be nonlinear or even nonmonotonic. In other words, it is not necessarily the case that a certain percent increase in neuronal activity is followed by an equal percent increase in vasodilation. Therefore linearity needs to be assessed on a case-by-case basis. In the case of PET responses, Fox and Raichle (1984) observed a nearly linear relationship between frequency of visual stimulation (within a range between 0 and 7.8 Hz) and changes in blood flow in primary visual cortex as estimated using O^{15}_2 -PET method (see Figure 35.5). The presence of linearity was considered critical for the application of “subtraction” methods during the analysis of hemodynamic responses.

If the linearity assumption is not met, it is still possible that the relationship between the dependent variable and the physiological phenomenon is monotonic. Although a monotonic relationship may limit the use of linear transformations and of parametric statistics,

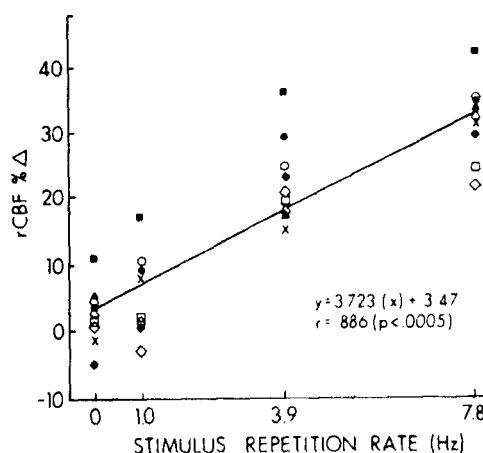


Figure 35.5. From Fox & Raichle (1984): the linearity of the relationship between stimulation frequency and blood flow measures obtained with O^{15}_2 -PET.

it is still sufficient for the application of nonparametric statistics based on qualitative judgments about effect sizes. Finally, relationships that are nonlinear, but follow a more complex function, are susceptible of transformations that can make them linear (e.g., the transformation of electrodermal resistance measures into conductance measures).

So far I have discussed the linearity of the relationship between the observed measures and the physiological parameter they intend to measure. To the extent that psychophysiological measures are used to make psychological inferences, it is also important to consider the type of relationship that exists between the physiological parameter and the underlying psychological concept being investigated. For instance, the purpose of measuring neuronal activity in a certain brain area may be the investigation of memory or attention phenomena, and the purpose of measuring blood pressure may be the investigation of the effect of psychological stressors on the cardiovascular system. Clearly, the issue of linearity (or at least of monotonicity) of the relationship is just as important here as in the case of the relationship between the observed measure and the physiological variable.

2.3. Continuous versus discrete measures

The distinction between continuous and discrete measures refers to the time at which information about the status of the physiological system under study is in principle available. Some psychophysiological measures can provide information at any time with no particular restriction. For instance, the diameter of the pupil can be monitored continuously, and provide a more or less continuous measure of the level of activation for particular subcomponents of the autonomic system. The EMG and EEG are continuous measures of the electric potentials produced by muscle and brain tissue, respectively. However, other physiological measures, such as measures of the heart rate

and respiration rate, reflect modification of parameters of cyclic events that occur at some intervals within the body. For these measures, information about psychological phenomena may be available only at discrete times. For instance, the state of activation of some subcomponents of the autonomic system can be studied using changes in heart rate, but information is available only when the heart beats (or when a particular EKG wave, e.g., the R-wave, is produced). Still other measures can only be obtained by using particular maneuvers that elicit special responses in the body. For instance, parameters of the blink reflex can only be measured by eliciting the blink reflex itself. Therefore, if the interest is in using the blink reflex to monitor changes in the status of a particular physiological system (e.g., one related to emotion), information may only be available at specific times (see Bradley, Cuthbert, & Lang, 1991). A similar situation is obtained when the P300 component of the ERP elicited by secondary task probes is used to monitor the level of attention of subjects to primary task stimuli (see Wickens, Kramer, Vanasse, & Donchin, 1983).

I will label those measures that can provide information about the state of a physiological system at any time “continuous,” and those measures that provide information only at specific times “discrete.” An advantage of continuous measures is that information about the time course of psychological phenomena is readily available on single trial records. Discrete measures can also provide information about the time course of psychological events, but their effective sampling rate is limited by the interval between the successive measurement windows. Finer sampling rates require pooling information obtained across trials, and are more complicated to obtain. An additional problem with some discrete measures is that the time at which information is available may not be completely under experimental control. For instance, certain measures related to circadian rhythms can only be obtained unobtrusively when the subject is awake. This may result in a “variable” sampling rate that may complicate the analysis. For example of solutions to this problem see Monk and Fookson (1986) and Monk (1987). Measures of heart inter-beat interval changes as a function of the time at which the stimulus is shown during a cycle can also present special problems of identification because the time at which measures are available (i.e., the next heart beat) varies depending on the phenomenon under study. Appropriate representation of the effects requires special techniques (see Jennings, van der Molen, Somsen, & Ridderinkhof, 1991).

3. SIGNAL EXTRACTION AND ENHANCEMENT

3.1. Domains of analysis

Psychophysiological data can be considered as elements of a multidimensional matrix. We can consider the following dimensions specific to psychophysiological measures:

(a) type of measure, (b) time of sampling (or time with respect to some anchoring event), and (c) location in space (comprising one, two, or three subdimensions). Other dimensions may also be considered, such as experimental condition, subject, group, and so forth, but these dimensions are not specific to psychophysiological measures. Because some of the dimensions are typically grouped together (such as the three spatial axes) it is useful to talk about domains of analysis (such as temporal domain and spatial domain).

For some, and perhaps most, experiments some of these dimensions may have only one cell and are, therefore, fixed. For instance, we may record only data from one location or for one psychophysiological measure. It is important, however, to consider that different values on each of these dimensions all refer to the same individual case, and, as a consequence, possibly may be correlated. Thus, psychophysiological data can be expressed as elements of a multidimensional measurement matrix:

$$M_{m,t,x,y,z,c,s} \quad (35.3)$$

Note that each element of the measurement matrix $\nu\chi\xi M \nu H\chi\xi$ has several indices corresponding to the type of measure used (m), the time of observation (t), the spatial coordinates of the observation (x , y , and z), the conditions of observation (c), and the subject from whom the measure is recorded (s). A graphic representation of the multidimensionality of psychophysiological data is presented in Figure 35.6.

The fundamental multidimensionality of psychophysiological data creates special problems in the statistical analysis. In fact, the probability of spurious observations (i.e., alpha error) increases with the number of observations, although typically not linearly. Whereas it is essential that appropriate controls are made to reduce the probability of reaching conclusions on the basis of spurious observations (i.e., rejecting the null hypothesis when it is true), it is also equally important to maintain an acceptable probability of reaching some conclusion in the presence of noisy data (i.e., to maintain adequate power).

An additional problem is that the large number of qualifiers (i.e., indices) used for each observation may sometimes make it difficult to generalize the results. For instance, if an ERP peak is observed at a latency of 400 ms in condition A and at a latency of 600 ms in condition B, should it be concluded that the same ERP peak is delayed in condition B compared to condition A or that an altogether different type of ERP activity occurs in the two conditions?

To address these issues it is usually convenient to perform some intermediate analytical (or signal processing) steps between the level of observation of the raw data and the inferential statistical analysis *per se*, in which the data are used to provide support for or against theoretical arguments. These preliminary steps were previously labeled “data reduction” or “quantification.” A consequence of

Multivariate nature of psychophysiological data

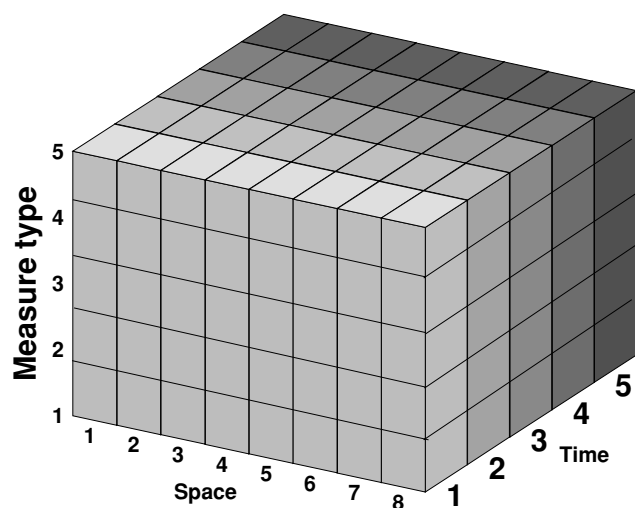


Figure 35.6. Graphic representation of the multidimensional nature of psychophysiological measures. Only three dimensions, specific to psychophysiological measures, are represented here (space, time, and type of measure). However, space itself may comprise several dimensions, and other dimensions (such as subject and experimental condition) are not represented. Note that the combination of different dimensions generates a large number of different observations for each subject. Because these are all observations from the same subjects, they may reflect different facets of the same phenomenon. Changes along one dimension may therefore be correlated to changes on another dimension.

these preliminary analysis steps is that the data should be no longer considered as “raw” observations, but as quantifications of parameters of some intermediate hypothetical construct that has heuristic value and is usually thought of as corresponding to some anatomical or physiological “entity” (e.g., activity in a particular brain structure in response to a particular stimulation condition or the blink response). In all cases, the data reduction implies some “interpretation” of the data. In other words, an analytical model is applied that makes more or less explicit assumptions about the underlying structure of the observed phenomenon, and involves some mathematical or logical transformation of the original “raw” data in order to estimate the value of some parameter of the underlying structure. The quality of the conclusions drawn from an experiment depends critically on the validity of these assumptions.

3.2. Temporal domain

3.2.1. Time domain and frequency domain

The time factor can be analyzed according to one of two basic models. According to the first model, the physiological events in the period under study evolve in a temporally ordered series that exhibits nonstationary properties (i.e., the basic structure of the activity changes over time). An example of data that fit this model well is the sequence of responses that occur in the brain after the presentation

of an individual stimulus. Another example is the changes in heart rate before and after the presentation of a critical stimulus. In these cases, it is profitable to use time as the main dimension of analysis (i.e., “time domain” analysis). Usually, particular aspects of the time series (such as maximum, minimum, integrated amplitude over an interval or a particular wave shape) are considered target features, and parameters of these features (such as amplitude, latency, or similarity with a particular template) are quantified and entered in the statistical analysis. In addition, time series can be averaged together (with or without alignment with respect to particular features) in order to increase the signal-to-noise ratio. This approach provides for a substantial data reduction, and data can be expressed in terms of properties of the target features. Averaging of ERP single trial waveforms and subsequent measurements of specific aspects of the waveforms (such as peaks) are exemplifications of this analytic approach.

The second model assumes that the basic structure of the activity does not vary over time and, therefore, exhibits stationary properties in which events repeat in a cyclical fashion. An example of data that appear to fit this model is the consistent response of the brain to a regular train of similar stimuli when habituation is not expected to play a role, and the oscillation in heart rate associated with respiration (sinus arrhythmia). When this model is appropriate, physiological activity can be analyzed by means of a set of tools that are able to extract the basic periodic structure of the activity – collectively referred to as “frequency domain” methods. In this case a new data series is built in which frequency replaces time as the ordering dimension, and data are expressed in terms of how much of the total variability over time is accounted for by fluctuations occurring at particular frequencies. Distinctive features of this new series can then be considered for quantification purposes (e.g., the “dominant” or most represented frequency in the data can be determined) in the same manner as time domain analyses. Frequency analyses may also provide information about the relative delay (or phase) at which particular rhythmic oscillations occur with respect to a reference time (such as the time of stimulation).

Recently, research has focused on mixed models, which assume that the presentation of a stimulus may lead to the *perturbation* of a basic oscillatory phenomenon. As a very simple example of this phenomenon, we may consider the change in the amplitude of the alpha rhythm (8–12 Hz) of the EEG that occurs when subjects open or close their eyes. Similarly, recent work has emphasized changes in gamma (approximately 40 Hz) and theta (4–8 Hz) rhythms that may accompany particular psychological processes (see Basar et al., 1999). Because it is assumed that a temporal event influences a specific rhythmic event, a mixed time-frequency model is required for the analysis.

3.2.2. Fourier analysis and autoregressive methods

Two types of analytic methods are most commonly used in the frequency domain: the Fourier transform and

autoregressive methods. Both approaches express the original time series in terms of how much certain frequencies are present in the data. However, the Fourier transform is based on a “deterministic” approach. That is, it represents all of the information (variance) contained in the original data in terms of the independent contributions of equally spaced frequencies. Autoregressive methods (of which there are actually different varieties) are instead based on a “modeling” approach. That is, they impose some constraints on the way in which individual frequencies contribute to the original time series and do not necessarily represent all of the variance in the data. This modeling approach reduces the number of “free” parameters in the data (i.e., it reduces the dimensionality of the data). This may allow the investigator to separate signal from noise.

The Fourier transform is a numerical method commonly used in engineering and other disciplines. A prerequisite for its application is that the sampling rate is constant across a given epoch (for an example of frequency analysis techniques with irregular sampling see Monk, 1987). The basic logic of this method is that any given time series (e.g., a recording epoch) can be expressed equivalently (i.e., without loss of information or interpolation/extrapolation) in the time and frequency domains. In the frequency domain a time series can be expressed as the sum of several time series, each characterized by the equal-amplitude oscillations of sinusoidal functions with frequencies equal to a multiple of the inverse of the length of the time series (e.g., if the recording epoch is one second long, then the frequencies used to describe it will be 1 Hz, 2 Hz, 3 Hz, etc.), plus an offset term related to the mean value across data points (usually called “0 frequency” or “DC value”). The number of frequencies is equal to the number of elements in the time series divided by two. So, if we are sampling a one second epoch at 128 Hz, the Fourier decomposition will be based on one DC value and 64 basic frequencies (1 Hz, 2 Hz, 3 Hz, ..., 64 Hz). Each basic frequency is associated with an *amplitude* value, related to how large the oscillation is at that particular frequency, and a *phase* value, related to the relative timing of the peak of the first oscillation with respect to the beginning of the time series. Derivation of the Fourier transform is numerically complex, and requires a number of exponential operations proportional to the square of the number of elements in the time series. Therefore, application of the Fourier transform to long time series may be cumbersome. Fortunately, when the number of elements in a time series is a power of 2 (e.g., 4, 8, 16, 32, 64, 128) the numeric problem simplifies and can be carried out using a procedure called the Fast Fourier Transform (FFT).

The outcome of the Fourier transform is a complex number associated with each frequency, in which the real and imaginary parts both contain information associated with the amplitude and phase of the basic oscillations. The real and imaginary components can be thought of as corresponding to the coordinates of a vector in a com-

plex plane. However, it is most convenient to express the results in terms of the polar coordinates of this vector (i.e., length and angle or orientation). This transformation separates the *amplitude* (equivalent to the length of the vector) and *phase* (equivalent to the orientation of the vector) associated with each frequency. Although informative, it should be remembered that this transformation is not linear. Therefore, data averaging (or other linear operations) are appropriately performed prior to this transformation.

These operations yield a set of phase and amplitude values for each frequency. In most cases, the attention of the investigators focuses on the amplitude data, although there are exceptions, such as in the study of visual evoked potentials (Tomoda, Celesia, & Taleikis, 1991). A common way of displaying the results is by plotting amplitude as a function of frequency (also known as the “periodogram”). In some cases, instead of the amplitude of each oscillation frequency, the data are expressed in terms of the square of this amplitude. This is called the “power” of a particular frequency. Amplitude and power are the frequency domain equivalents of the standard deviation and variance in the time domain. Indeed, the sum of the power values for all frequencies is related to the total variance in the data. For this reason, the Fourier transform can also be used as a method for partitioning the variance observed in a time series into subcomponents with different frequencies. As a consequence, it is possible to apply some of the standard tools used for analyzing variance (e.g., chi square test and analysis of variance) to power frequency analysis. Finally, it is a common practice to pool together the power for an interval of frequencies within a certain range (i.e., a frequency band). For example, in an EEG study we may be interested in quantifying the total power for frequencies ranging between 8 and 12 Hz (usually called the alpha band) and in a heart rate study on sinus arrhythmia we may wish to quantify the power for frequencies between 0.2 and 0.5 Hz (i.e., the respiration frequency range).

As mentioned above, the Fourier transform is based on the observation that any time series can be decomposed in a number of sinusoidal waves that extend with equal amplitude across the entire time series. Although mathematically valid in all cases, this way of describing the data may, in some cases, produce results that require further interpretation. This occurs for several reasons.

First, there may be cases in which the oscillations in the time series can be best described by functions other than a sinusoid. For instance, the pressure waves that propagate through the arteries after a heart beat are, at first approximation, triangular in shape (see the upper portion of Figure 35.7). When the shape of the oscillations departs from sinusoidal, the Fourier transform will decompose each wave into subcomponents (called “harmonics”), with each subcomponent having a frequency that is a multiple of the basic frequency (called “fundamental”) at which the wave repeats. As an example of this decomposition, the periodogram of the saw-tooth arterial pressure waves is reported at the bottom of Figure 35.7. Note that the

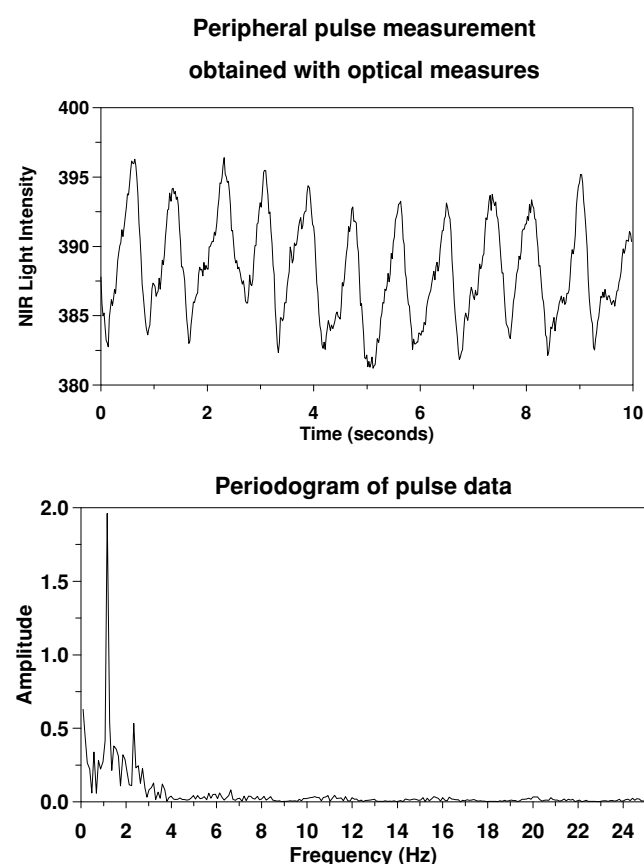


Figure 35.7. *Top:* Recording of the peripheral pulse from one subject during a psychological experiment obtained with near-infrared (NIR) optical methods (see Gratton, 1997). Note that intensity of the NIR light drops rapidly approximately every 0.8 seconds. The drops correspond to the times at which the arterial blood is pushed into the blood vessels by the beating of the heart. Each drop is followed by a slower increase in the light intensity, which reflects the flow of the arterial blood out of the blood vessels. The quick drop in light intensity, followed by the slow increase, generates a triangular shaped ("saw-tooth") wave that repeats approximately every 0.8 seconds. *Bottom:* Amplitude spectrum ("periodogram") corresponding to the data reported in the top portion of the figure is obtained using an FFT method (the DC component is omitted). Note that the spectrum is dominated by a 1.2 Hz frequency, which corresponds to the heart beat frequency. However, activity is also evident at 2.4 Hz and at 3.6 Hz, which are the first and second harmonics of the fundamental 1.2 Hz frequency. The presence of activity at these frequencies reflects the fact the cyclical pattern is not sinusoidal.

periodogram indicates activity not only at the basic frequency (around 1.2 Hz) but also at some of its harmonics (such 2.4 Hz). The pattern of harmonics is related to the particular wave shape of the basic, cyclical pattern. Given that in most psychophysiological measures the basic wave shape is not a sinusoid, psychophysiologicals are often interested in the patterns of the harmonics as much as in the fundamental frequency.

A second way in which data may depart from the description underlying the Fourier transform is when the type of activity observed in one part of the epoch differs substantially from activity in another part of the epoch. An

extreme case is that of "phasic" features, which appear only once in a time series. Phenomena of this type, which are observed frequently in psychophysiological data, may be difficult to study using Fourier-transformed data because the variance associated with them will be distributed in a very complex manner between the DC component and various frequencies. However, this variance will still be represented in some form in the Fourier transform. In other words, the partitioning of the variance obtained using Fourier transforms is best suited for phenomena that repeat cyclically along the time series (i.e., "stationary" waves).

3.2.3. Mixed time-frequency models

3.2.3.1. Joint time-frequency domain analysis. In recent years, combinations of time-domain and frequency-domain approaches have been introduced (see Stiber & Sato, 1997). The major justification for these hybrid techniques is the assumption that the way in which the brain processes information involves the rapid modulation of basic frequencies of activity. A frequency that has been considered as particularly important in brain activity is approximately 40 Hz (also called the "gamma" band; see Gray, Engel, König, & Singer, 1989). According to this view, information processing may involve the amplitude modulation of a 40-Hz activity. When a new stimulus is presented, the oscillations at this frequency increase in size. A converse phenomenon is in some cases observed for the alpha band ("alpha blocking" or "enhancement"; see Pfurtscheller & Neuper, 1992). Other examples include (a) the analysis of the modulation of sinus arrhythmia over time as a method for studying the time course of vagal activity, and (b) the analysis of EMG activity, which often involves the occurrence of a brief series of high-frequency oscillations (typically above 30 Hz).

In all these cases, although the psychological events are transient (and therefore evolve over time), their physiological manifestations involve the modulation of a cyclic phenomenon. To represent appropriately this modulatory phenomenon it is necessary to study the change over time of the amplitude of specific frequencies. This involves a segmentation of the time series in shorter epochs, the determination of the amplitude of the frequency of interest in each epoch (the analysis of which is performed in the frequency domain), and the analysis of the time course of modulation of the amplitude of the frequency of interest (the analysis of which is performed in the time domain). This procedure is labeled Joint Time-Frequency Analysis (or JTFA; see Tallon-Buadry, Bertrand, Delplanche, & Pernier, 1996). In the case of EMG, the response is often analyzed entirely in the time domain, and involves: (a) recording EMG data using a relatively high-frequency band-pass (10–100 Hz), (b) rectification of the waveforms (i.e., transformation of the negative values into positive values), and (c) low-pass filtering (typically below 20 Hz). This process is adequate for determination of latency measures, but requires recordings with low noise levels.

3.2.3.2. Wavelet decomposition. Another method combining features from both time- and frequency-domain analyses is the wavelet decomposition method (Samar et al., 1999). Similarly to the Fourier and autoregressive methods, this procedure is also based on decomposing a complex waveform into subcomponents (called wavelets), which when added together form the original waveform. However, different from the frequency-domain methods, it is not assumed that these basic wavelets are cyclic or stationary. Rather, they are allowed to vary in duration and latency as well as amplitude. This “modeling” effort can be used to reduce the dimensionality of the data and fit an underlying structure to a particular waveform. This approach can be particularly useful for single-trial analysis of EEG data.

3.2.4. Amplitude domain measures

All of the temporal-domain data processing steps outlined so far result in descriptions of the observed activity as a function of time, frequency, or a combination of both. However, in some cases, it is preferable to analyze the overall “amount” of observed activity, without expressing it as a function of time or frequency (Cacioppo & Dorfman, 1987). This may be appropriate when the time or frequency variable is considered irrelevant, or when variations of the measurement over time depend on factors that are not related to the physiological (or psychological) variable of interest. For instance, when the stimulation condition cannot be conveniently subdivided into specific time epochs or periodic events, it may be advantageous to consider the overall physiological response observed during an experiment, rather than considering the individual responses. An example of this is the number of electrodermal responses during the presentation of a particular videotape that is not characterized by clearly identifiable time markers. Another case is when measures cannot be taken repeatedly, or whose time of acquisition is longer than the time during which the experimental manipulations are performed. An example of this is deoxyglucose-PET data, in which measures of the amount of radioactive material accumulated in various brain areas are taken at the end of the experiment (Cherry & Phelps, 1996). The temporal dimension in this case is reduced to one data point, and absolute amplitude measures are taken. Finally, there are cases in which there is interest in summarizing a unique number for all the variability of observations obtained across an extended epoch (e.g., Elui, 1969; Dorfman & Cacioppo, 1990).

3.3. Spatial domain

3.3.1. Spatial dimensions

In recent years there has been a substantial growth of interest in the spatial characteristics of psychophysiological measures because they may correspond to the anatomy of the human body, particularly to the brain. A particular impulse for this enterprise has been given by the intro-

duction of functional neuroimaging methods, for example, single-photon emission computerized tomography (SPECT), near-infrared spectroscopy (NIRS), PET, fMRI, and EROS. Brain anatomy is not conveniently expressed using frequency-domain methods because of the fundamental uniqueness of the various structures, thus other methods of analysis are most commonly used. An exception, though, is given by the use of the Fourier transform approach in the decoding of spatial information in MRI. (see Cohen, 1996).

Psychophysiological measures differ in the number of spatial dimensions used, and in their relationship with underlying anatomy. Some measures refer to brain activity (e.g., EEG, ERPs, MEG, fMRI, PET, NIRS, and EROS), whereas others refer to the activity of other body structures (e.g., EKG and heart rate, EMG, pupillary diameter, measures of the blink response, measures of EDA, blood pressure, and respiration). Although measures of both brain activity and bodily functions can be taken at different locations, usually the second group is less directly dependent on brain anatomy than the first.²

All measures of brain activity produce substantially different results depending on the location at which they are taken. Some of these measures provide data that can be localized to specific brain volumes (e.g., PET, fMRI, NIRS, and EROS). For these measures any recorded value will have three indices, which refer to the x, y, and z spatial coordinates of the observed value. Other measures (e.g., ERPs and MEG) provide data that refer to surface observations and have, therefore, only x and y coordinates. For these measures algorithms have been proposed that allow for the reconstruction of depth information, and thus pinpoint brain structures that are responsible for the activity observed at the surface. However, these reconstruction algorithms require a modeling effort that, for both ERPs and MEG, has multiple solutions, which are equally valid from a statistical point of view. Therefore, limited confidence must be placed on the results.

Maps and tri-dimensional (3D) reconstructions obtained with various techniques can be evaluated on the basis of two major properties: (a) localization power (i.e., the ability to attribute a particular activity to a particular location), and (b) resolution (i.e., the ability to distinguish activity coming from two closely located points). These two properties are different and should not be confused. Some techniques may possess a very good spatial localization power, but limited spatial resolution. This occurs because, as is often the case, the larger the activity, the greater the space over which it can be detected

² There are, however, notable exceptions to this rule, in particular in the area of the study of reflexes. For instance, Hackley and Johnson (1996) have used properties of the blink response (a peripheral measure) to determine whether a particular phenomenon (pre-pulse inhibition) is under the control of cortical or sub-cortical structures (which is clearly an issue of localization of function within the brain). This indicates that, if appropriate experimental designs are used, peripheral measures can provide some information about the localization of function within the brain.

(even from outside of the location at which it is generated). Therefore, two closely spaced sources may end up being fused in one single “blob.”

3.3.2. Analysis of surface data: Surface maps and 3D inference

As previously mentioned, some psychophysiological measures (ERPs and MEG) can be used to generate maps of the distribution of a particular physical parameter at the surface of the scalp. Usually, these maps are generated using a process of interpolation: a reduced number of surface observations, usually less than 50, but in some cases as many as 256, are used to derive representations of the activity over the whole head, or a segment of it. The interpolation procedure requires the assumption that the spatial sampling used represents the surface distribution of the underlying phenomenon in a satisfactory way.

When considering the time domain, an issue arises concerning the minimum spatial sampling required for producing representative electrical (and magnetic) maps of brain activity. In general, information theory indicates that the spatial resolution, but not necessarily the localization, is limited to twice the distance between two adjacent measurement locations. This is called the “spatial Nyquist frequency.” At first glance, it may appear that a greater spatial sampling will result in maps with superior quality. Some investigators have shown that the information content increases with the number of recording locations (Srinivasan et al., 1996). Their results make intuitive sense. However, practical and theoretical problems may limit the advantage of increasing spatial sampling of ERP and MEG data. One limiting factor is the cost of the devices required to record and analyze data from a large number of channels. Another limiting factor is that measurement errors are likely to increase with the number of recording channels, which may effectively decrease the signal-to-noise ratio.

To illustrate this problem, let us suppose that an experimenter is interested in running an ERP study to determine where a potential reaches its maximum over the scalp. Let us also suppose that, in this experiment, there exists a .01 probability that a large, inaccurate value (i.e., an artifact) may influence the measures at any given electrode location, and if such an artifact occurs, the maps may identify a maximum at an erroneous location. If only a small number of electrodes are used (e.g., <10), the probability of a “false” maximum is relatively small ($p < .10$). Some confidence can be attributed to the results, although the spatial resolution will be very low. However, as the number of recording locations increases, the probability that at least at one location there will be an artifactual value also increases. Therefore, the probability of incorrectly localizing the maximum will also increase, and, if enough electrode locations are used (e.g., over 100) the probability of an incorrect localization would become very high ($p > .63$). In this case, little confidence could be attributed to the results. This emphasizes the need of

using the “appropriate” number of recording locations. It also emphasizes the difference between spatial resolution and localization – procedures that enhance spatial resolution do not necessarily enhance localization, and vice versa.

As mentioned earlier, most maps are constructed using an interpolation process. There are various algorithms for interpolations. A relatively simple procedure involves using a linear interpolation to derive the values in between the observed locations. A problem with this method is that the maximum of activity in the maps is bound to be at one of the locations at which the recordings were obtained. However, more sophisticated techniques are now commonly used. Some of them are based on fitting polynomial surfaces, or other types of surfaces obtained with some form of smoothing function, through the various observed data points (e.g., spline interpolation; Perrin et al., 1987). These interpolations may increase the localization of activity in surface maps, but at the cost of a reduced spatial resolution. In addition, assumptions need to be made about the spatial frequencies contained in the data.

Some transformations of the surface maps may be useful for counteracting some limitations of the original observations. For instance, limitations of surface maps of the electrical potential over the scalp include: (a) the absolute value at each data point is dependent on the choice of reference, and (b) activity tends to be spread out across extended areas. This may make it difficult to appreciate differences between conditions. A way of addressing both of these issues is transforming the measures mapped from voltage difference into electric currents (“current source density,” or CSD, maps). According to Ohm’s law, currents flowing between two points are directly related to the differences in potential and inversely related to the resistance between the points. If the resistance is assumed to be constant, then current maps can be easily computed from voltage maps. In the case of the scalp, the current can be considered to flow almost exclusively along the skin, which is a path of lower resistance than the skull underneath. The flow of current in two dimensions can then be described as a current density, which is computed using the 2D spatial derivative of the voltage map. In practice, this provides a measure of a local gradient, which is independent of the definition of the reference level because a local reference system is used, and highlights local details of the maps. However, phenomena that extend across large areas become less visible. Furthermore, small error variance may be amplified by this method.

Whereas CSD maps (and other transformed maps) of surface potentials may be useful for highlighting differences between conditions, they should not be interpreted as 3D reconstructions of the activity inside the head. Such a reconstruction requires a model of (a) how the signal is generated inside the head, and (b) how it is transferred from interior sources to the surface. Both problems require different solutions depending on the type of

measure adopted. For electrical activity (EEG and ERPs), the propagation of the potential to the scalp requires information about the conductive properties of various media interposed between the sources and the surface electrodes (such as gray and white matter, cerebrospinal fluid, bones, and skin). Furthermore, because of the relatively low frequency of the data, impedance is typically not considered an issue here. The transmission of magnetic fields is less influenced by departures from homogeneity in the media than that of electric fields. Until recently, all source analysis algorithms were based on abstract models of the head (e.g., the “three-sphere model”), which modeled the head as a simple geometric shape. In the last few years, realistic head models, which are derived by using MRI or computerized axial tomography (CAT) scans, have been adopted, and the propagation of activity to the surface has been computed using finite computation methods. However, knowledge about the *in vivo* conductive properties of head tissues is still imperfect and may create substantial errors in the computation.

Notwithstanding these problems, substantial developments have been made in the modeling of the sources of electrical and magnetic activity in the last 20 years. Three types of approaches can now be used:

1. Regional source models. This approach models the propagation of potentials or fields within the head, rather than their generation. It produces results that tend to be quite robust (i.e., it converges to similar results if small variations in the observed activities are compared), but has low spatial resolution (up to several cm; Pascual-Marqui, Michel, & Lehmann, 1994).

2. Diffuse source models. With this approach, it is usually assumed that activity is generated by a large number of sources, for instance, the entire cortical surface. To reduce the number of free parameters, some assumptions can be made, for example, that sources are normal to the cortical surface, have overall as little activity as possible (i.e., minimum norm), and are spatially smooth. Diffuse source models may possess intermediate spatial resolution, but their accuracy is difficult to evaluate because they are typically largely overdetermined or the number of free parameters to estimate is several orders of magnitude greater than the number of observations (Dale & Halgren, 2001).

3. Point-dipole models. These models can be taken at a particular time point, or combine information accrued over time, which generates “spatiotemporal” dipoles (Scherg & von Cramon, 1986). This approach has greater spatial resolution and localization power, but also a greater level of uncertainty about the number of modeled sources. By and large, source modeling of electrical activity and, to some degree, magnetic activity still requires external validation.

The problem of how to derive 3D reconstruction of data from data with lower dimensionality is not restricted to

EEG, ERPs, and MEG, but is true for all imaging techniques. Each of these imaging methods solves the problem in a different way. CAT and PET scans use a large number of surface detectors to infer activity inside the brain. Both of them are based on high-energy radiation (photons) with minimum scattering. They use a form of 3D reconstruction that is based on inferring the distribution of activity inside the head from surface activity, which ignores variations in the local anatomy, but is entirely based on geometric and statistical principles.

Optical imaging methods (e.g., EROS) use light wavelengths at which scattering is the dominant phenomenon; further diffusion parameters (scattering and absorption coefficients) vary conspicuously between tissue types. Thus, a 3D reconstruction of optical imaging data needs to take into account these properties of the medium, similarly to EEG and ERPs. However, because the movement of photons inside the head is strongly limited by the scattering and absorption properties of the head, the spatial resolution of optical imaging is inherently much higher than that of methods based on electric potentials and magnetic fields.

Magnetic resonance imaging methods use a completely different approach to 3D reconstruction. Different from the instruments for electrophysiological, optical, and PET recordings, MR machines typically have one, or very few, detectors. In this case, to provide spatial information, the magnetic fields used to induce and measure the magnetization of individual nuclei are varied over time and space in a very special manner so to provide “tags” indicating where the signal originated. Whereas this technology provides, in principle, a very high spatial resolution, it generates a few problems: (a) there are limitations in how quickly the signal can be acquired, and (b) it is essential that the local variations in the magnetic fields are accurately predictable. To partially address this second issue, it is customary to “shim” the magnets before recording data. Nevertheless, characteristic artifacts are present when distortions in the magnetic fields occur because of, for example, the presence of metal prosthesis and dental work.

3.3.3. Analysis of 3D spatial data

Imaging methods provide 3D descriptions of how the signal is generated inside the body. If the data refer to changes in the strength of the signal as a consequence of the functioning of a particular organ, they are called “functional imaging methods.” Examples of functional imaging methods include fMRI and PET. Functional imaging methods vary in terms of spatial resolution and localization power, as well as other dimensions (e.g., how direct the measures are and how much temporal resolution they possess). The localization power of some of these techniques constitutes their major advantage, but it also requires attention to a number of methodological issues. One of these issues is that the anatomy of some organs – the brain in particular – varies across individuals. Therefore, a signal

that may appear at a particular location in one person may appear at another location in a different person. In order to generalize conclusions across people it is indispensable to “align” the data obtained in different individuals. This process requires re-scaling the various features of the anatomy of each single person to a “standard” anatomy, so that phenomena observed in different people can be directly compared.

A commonly used procedure to align different brain anatomies to a standard anatomy was introduced by Talairach and Tournoux (1988). According to this procedure, structural scans of brains of different individuals, which can be obtained using MRI or CAT scans, are first aligned using a set of basic anatomical features. The brain is then subdivided into regions, and each region is re-scaled independently to the dimensions of a standard provided by the Talairach and Tournoux atlas. This allows researchers to compare data across different subjects, in order to examine the reliability of findings, and compute “average” brain maps across subjects. Using the standard atlas, it is then possible to associate specific functional data to particular anatomical structures of the brain (e.g., the Brodmann cytoarchitectonic areas of the cortex). This procedure is likely to introduce small distortions because the large subdivisions used in the Talairach and Tournoux system may not capture all of the individual differences in anatomy. Furthermore, the subdivisions are based on subcortical, rather than cortical landmarks. Because the cortex is the most variable part of the human brain, the Talairach and Tournoux system can sometimes lead to significant alignment problems. Furthermore, the Talairach and Tournoux method is based on a single “model” brain. A variation of this approach is the Montreal Neurological Institute (MNI) model, which is based on a statistical average of a large number of brains. However, the Talairach and Tournoux transformation is a standard method that can provide useful approximations in a number of cases, and allows scientists to compare results obtained in different experiments and labs.

More recently, other alignment methodologies have been introduced, which take into account cortical features. Some of these methods preserve the 3D structure of the brain, whereas others generate flattened images of the cortical surface (Van Essen et al., 1998). These methods are generally based on complex warping algorithms in which each location on the surface of the cortex of a particular brain is associated, because of its relative position with respect to local landmarks, with a point in a standard 3D or flat representation of the cortex.

An additional problem with 3D spatial data is the huge number of dependent variables used. A 3D scan of the brain based on voxels (i.e., the small volumes used for sampling a large volume) with a 1.5–3 mm side may generate more than 100,000 data points, each of which should be considered as a dependent variable. This has two major consequences. First, multivariate techniques are difficult to apply to these data, unless a strong reduc-

tion in the number of points is considered. Second, even univariate analyses need to take into account the problem of multiple comparisons (i.e., the inflated probability of alpha error when a large number of comparisons are carried out). There are various solutions to this problem. One is to carry out analyses only for selected points, or regions, labeled regions-of-interest (ROIs). This, however, restricts the analysis to small portions of the brain. A second, simple solution to this problem is to apply the Bonferroni inequality, which consists in setting the criterion of alpha error for each comparison to $crit/nc$, where $crit$ is the criterion normally used (e.g., .05) and nc is the number of comparisons (in this case, data points). This guarantees that the overall probability of alpha error is less than the original criterion. However, if the number of data points, or voxels, is large (e.g., 100,000) only data points whose statistical values are associated with extremely low probability values will be significant. Although this analysis would protect against alpha error, it would reduce the power of the study as to make it virtually useless.

A solution to this problem has been proposed by Friston (1996). This solution is based on the application to neuroimaging of random field theory (also called “blob” analysis), which was developed to deal with similar problems in other disciplines. Blob analysis is based on the idea that the observations obtained at different locations are not independent from each other – close voxels are likely to be similar to each other. This confers some level of “spatial smoothness” to the imaging data, and implies that the number of independent comparisons is in fact much smaller than the number of voxels recorded. In other words, although there may be 100,000 voxels recorded, there may be 1,000 or fewer independent comparisons. To determine what statistical level is needed to reach significance, it is important to establish the spatial smoothness of the data. A critical point is to establish the *spatial smoothness of the error term* and not that of the effect to be studied. This can be achieved in various ways. The most effective is based on the computation of the spatial derivative of the error term. This approach to the analysis protects against alpha errors and still maintains a reasonable power.

An interesting consequence of the use of spatial smoothness in the estimation of the statistical significance of the data is that the greatest power is obtained for data with high spatial smoothness. However, spatial smoothness can be considered as the reciprocal of spatial resolution – the higher the spatial resolution of the data, the smaller the spatial smoothness. Thus, in order to increase the power of a particular brain imaging technique it may be advantageous to reduce its spatial resolution. This may be easily achieved by “spatially filtering” the data (see section 3.4), so that data contain only relatively low spatial frequencies. In fact, most brain imaging studies use spatial filters that reduce spatial resolution to the order of 1 cm or less.

3.4. Filtering and artifact problems

One of the major issues in processing any type of data, in particular psychophysiological data, is that of distinguishing the signals of interest from noise. Noise can be broadly defined as any other phenomena observed in the data apart from the signal(s) of interest for the investigator. In general, the ability to distinguish signals from noise can be quantified using a construct called the “signal-to-noise” ratio, which is the amplitude ratio between the signal and the noise. The larger the signal-to-noise ratio, the easier it is to identify the signal. As a consequence, more reliance can be attributed to the observations. One of the major tasks of signal processing is to increase the signal-to-noise ratio. This can be achieved by amplifying the signal, reducing the noise, or both. In general, procedures or devices that reduce the amount of noise present in the data are called filters.

By and large, filters are based on the principle that the signal can be distinguished from the noise on the basis of some characteristic features. For instance, the signal may have a particular frequency that is different from that of the noise. Therefore, by amplifying the frequencies carrying the signal and dampening the frequencies carrying the noise, it is possible to increase greatly the signal-to-noise ratio. This is common practice in psychophysiological recording, and can be achieved while the data are recorded by the use of online, usually analog, filters or at any stage of data analysis with offline, usually digital, filters.

Generally filters are described in terms of the frequency at which they attenuate the signal. High-pass filters cut off activity with a frequency lower than a designated frequency, whereas low-pass filters cut off activity with a frequency higher than a designated frequency. Other types of filters, called notch filters, only cut off a very specific frequency (e.g., 60 Hz), and leave activity with lower or higher frequencies unaltered. Usually, the cut-off point of a filter indicates that at that point activity is reduced by 3 dB (approximately 70%). However, other parameters are important in describing the performance of a filter. Specifically, the performance of a filter is described by a “performance operating characteristic” (POC) function, which describes the proportional attenuation of the signal at different frequencies. Ideally, a filter should maintain all of the activity up (for low-pass filters) or down (for high-pass filters) to the cut-off frequency and eliminate all of the frequency above or below the cut-off frequency.

Most psychophysiological recordings involve online analog filters. The advantage of these filters is that they can reduce high frequencies before digitization occurs. This may reduce the possibility of aliasing high frequencies. As already mentioned, aliasing occurs when digitization is performed at a rate that is less than twice that of a frequency present in the data. In this case, activity at a high frequency will be reflected into activity at

a lower frequency. Because aliasing cannot be eliminated once it occurs, it is very important to prevent it by filtering high frequencies before digitization. Another advantage of online filters is that they can be used to cut-off large noise oscillations, which usually occur at very low frequencies. These large oscillations may generate signals that are outside the operating range of the analog-to-digital converter, which will “saturate” (i.e., it will provide an output that corresponds to the maximum or minimum value allowed by the converter, but not to the actual value).

Online frequency filters, however, are usually recursive: filtering is based on the activity that has already occurred and not on the activity that has yet to occur. Therefore, they introduce a phase distortion in the data – phenomena may appear to peak at different times from when they actually do, or their shape may be distorted.³ Online low-pass filters, which cut off high frequencies, tend to displace peaks to a later time than when they really occur. Online high-pass filters, which cut off low frequencies, tend to displace peaks to an earlier time than when they really occur. Offline filters can be designed to be nonrecursive and, therefore, do not necessarily produce phase distortion. The extents to which these displacements occur depend on the frequencies that are cut off by the filters. It is, therefore, preferable that data be recorded online using a wide band-pass, and that filtering used for signal enhancement be mostly carried out offline using nonrecursive filters.

Various types of offline digital filters are available. Because they do not need to be analog, it is relatively simple to produce filters with very good performance (i.e., with very sharp cut-off points). Whereas filters with such characteristics may be built using a frequency-domain transformation of the data, the operations required are complex and may slow down the data processing exceedingly. It is possible to construct filters that work in the time domain that approximate the performance of these filters. Cook and Miller (1992), as well as Farwell, Martinerie, Bashore, Rapp, and Goddard (1993), present a review of some of these filters.

Wiener (1964) introduced the concept of “optimal” filter. This is a filter whose POC function is optimized to increase the signal maximally compared to the noise. For instance, a filter may be built to maximize the between-condition variance with respect to the within-condition variance. To build such a filter it is necessary to define the signal in some manner, so that the frequency carrying the signal can be identified *a priori*. In addition, an optimal filter is practical only if it is designed in the frequency domain. However, as already mentioned, filtering in the frequency domain may actually reduce the speed of signal processing considerably. These problems have limited the use of optimal filters.

³ This phenomenon can be minimized by using a special type of filter (e.g., elliptical and/or Bessel filters).

In certain cases the investigator is interested in studying activity that is characterized by very specific frequencies. For instance, if the investigator is interested in the response to stimuli presented at regular intervals, the investigator may assume that the response also should be observed at regular intervals. In this case, the frequency of the signal is well known – either the frequency of stimulation or one of its harmonics – and activity at all other frequencies may be discarded as noise. An “inverted-notch” filter (also called a “band-pass” filter) may be used to selectively analyze the response, and may enhance greatly the signal-to-noise ratio. Procedures of this type are used to analyze “steady-state” evoked responses (Tomoda et al., 1991).

So far, this discussion has focused on filters used in the temporal domain to eliminate activity at undesirable frequencies. However, filters can also be built in the spatial domain in order to amplify activity with certain, particular spatial properties with respect to other types of activities. An example of this is the Vector Filter procedure proposed by Gratton, Coles, and Donchin (1989). The Vector Filter procedure can be considered as analogous to a planned contrast executed in the space domain. Gratton, Kramer, Coles, and Donchin (1989) and Fabiani, Gratton, Karis, and Donchin (1987) showed that this procedure could help in ERP component identification. This notion has been recently re-introduced with the label of “virtual channel,” which refers to a particular EEG channel constructed by appropriately weighting a number of different channels. The weights can be selected on the basis of specific procedures, for example, Principal Component Analysis (PCA; Spencer, Dein, & Donchin, 1999), independent component analysis (ICA; Makeig et al., 1997), or other methods.

A special type of noise, present in most psychophysiological data, is referred to as an “artifact.” This term is normally used to indicate large, isolated noise activities that originate outside the system of interest. Examples of artifacts include (a) the potentials associated with blinks and other types of eye movements during the recording of brain electrical activity, (b) susceptibility and movement artifacts during MRI and PET recordings, (c) missed heartbeats in measures of heart rate, and (d) pulse artifacts in the EROS and NIRS data. Artifacts may be very large – several times larger than the signal – and could, therefore, completely obliterate the signal. Furthermore, they may occur in a systematic fashion (i.e., the frequency of artifacts may be consistently greater in one experimental condition than another). However, artifacts are often easy to recognize because they have specific “signatures” that are clearly distinguishable from regular data. For instance, blinks generate electrical potentials with a characteristic spatial distribution over the scalp, and missed heartbeats in heart rate measures can produce exceedingly long intervals that are outside of the normal variability.

A number of manual and automatic procedures have been developed to detect, eliminate, or compensate for the effects of artifacts. These procedures vary from a particular physiological measure to another and, therefore, will not be reviewed here. It is important, however, to remember that appropriate procedures for dealing with artifacts are an essential step in the processing of any physiological signal.

4. DATA REDUCTION AND QUANTIFICATION

4.1. Quantification

Practically all analyses of physiological measures require a step in which a small number of numerical values are extracted from the large amount of data recorded, and are then used for inferential statistics. This step may be called quantification. Quantification usually involves two steps: (a) identification of a particular feature considered to represent a given physiological event, and (b) measurement of some parameter of this feature.

4.1.1. Feature identification

The procedures used to identify the feature of interest are quite variable. In some cases the feature can be easily identified by visual inspection of the data. This occurs when the signal-to-noise ratio is so large (e.g., >3:1) that the feature of interest can easily be distinguished from sources of noise or from other signals (e.g., blinks in electrooculographic (EOG) recordings, and R-waves in the EKG). In other cases the features of interest are buried under variable, and sometimes large, amounts of noise or other signals. An example of this is given by various components of the ERP, especially when measured from single trials, or from averages of a small number of trials. In this case, identification of the component of interest may be very complex, and require sophisticated pattern recognition algorithms, filtering procedures, or a combination of both. Thus, identification of the P300 component of the ERP on single trial recordings may require using a relatively heavy low-pass filter (band-pass 0–6 Hz or less) and pattern recognition algorithms such as cross-correlation methods (Fabiani et al., 1987; Gratton, Kramer et al., 1989). A step that is often required for identifying ERP components is the definition of a particular “time window” in which the component is expected to appear. Thus, it is possible to define P300 as a positive peak in the ERP waveform appearing in an interval between 300 and 700 ms, whereas a peak appearing earlier or later may be classified as a different component. Other attributes may also be considered important, for example, the particular location at which the activity is observed.

4.1.2. Measurement

The second step of the procedure is the measurement of a particular parameter of the feature of interest. Again,

parameters vary along different dimensions. By and large, three dimensions are considered in most cases: (a) a temporal dimension, (b) a spatial dimension, and (c) the intensity or frequency of occurrence.

4.1.2.1. Temporal dimensions. In the case of temporal dimensions, activity can be considered as having a particular latency, frequency, or phase. Measures include onset, peak, center-of-gravity, and fitting of particular functions to the data. The two measures that are most commonly used are onset and peak latency.

Onset measures may be very informative because they may define the latency of a particular physiological event as well as the maximum latency of the psychological phenomenon that is intended to signal. Onset can be measured as the first data point in the time series exceeding a preset value. This value can be obtained from previous work or from statistical computations of the variability in the measurements (Miller, Patterson, & Ulrich, 1998). However, the exact onset time of physiological measures can be difficult to determine when the measures contain even low levels of noise. In case of noisy measures (e.g., ERP activity measured on single trials or on a small number of trials), onset of a particular activity can be estimated by fitting regression lines to different segments of the data (Barrett, Shibasaki, & Neshige, 1986), and by determining the point at which the regression line corresponding to the pre- and post-onset periods meet. An alternative procedure is to consider the “half-amplitude” latency, which is the latency required for the signal to reach half of its maximum peak; other proportions of the maximum value can also be used (Smulders, Kenemans, & Kok, 1996). Whereas the latter approach is quite reliable (Smulders et al., 1996), the determination of which proportion of the maximum value to use is arbitrary, and the measure may not correspond to any particular psychological (or physiological) phenomenon. In addition, if the maximum value is systematically different across conditions, determination of the latency of half-amplitude measures may be misleading. An example is presented in Figure 35.8.

Peak measures are obtained by selecting the data point (in the time or frequency domain) of the time series at which the measure reaches its maximum value. If several peaks are present, the peak point can be defined as the maximum value within a certain interval, or as the “n-th” peak in the waveform (i.e., the first peak, the second peak, and so on). Peak measures tend to be more reliable than onset measures. However, they may be very sensitive to high-frequency noise. It is, therefore, often advisable to apply a low-pass filter before estimating the peak. The main advantage of peak latency measures is that they are easy to take even on noisy data. A disadvantage, in comparison to onset measures, is that they do not necessarily identify a point in time that is of theoretical importance. Thus, although from the onset of a particular activity we may infer that a given psychological process must

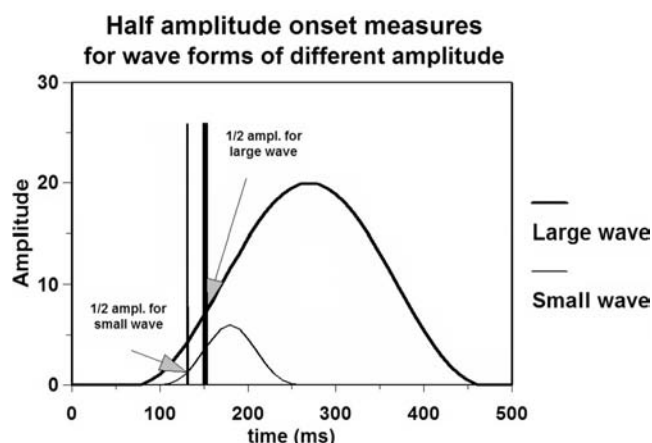


Figure 35.8. Half-amplitude onset measures for waves of large (thick curve) and small (thin wave) amplitude. The half-amplitude latency of the large wave (thick vertical line) was 170 ms, whereas that of the small wave (thin vertical line) was 147 ms. If half-amplitude measures are used to estimate onset latency, it may lead to the incorrect conclusion that the onset of the large wave occurred after the onset of the small wave.

have been carried out for the activity to occur, the peak of the physiological response may occur some time later and have no specific meaning. An extreme example is given by the time course of the hemodynamic response (measured with fMRI) and following the presentation of a train of stimuli. The time at which the fMRI response reaches its peak may provide little information about the time course of psychological events. The onset of the fMRI activity in a particular brain area, however, can be considered as the upper limit for the latency of activation of that area. It should be noted, of course, that activity in the area might in fact begin some time before it is detected. A similar situation exists for the latency of the peak of the skin conductance response. Note, however, that although for both types of signals the peak latency may provide little information about the timing of psychological events, the onset and the amplitude of either response may provide useful information about the nature of psychological events.

In general, temporal dimensions are considered as interval measures and are most commonly analyzed using parametric statistical approaches. However, temporal measures have usually a limited range. Additionally, in some cases, the distribution of the measurement error departs substantially from normality and both skewness and platykurtosis may be present (Fabiani et al., 1987). Both of these problems may lead to reduced power for statistical analysis.

4.1.2.2. Spatial dimension. The spatial dimension is most commonly analyzed in terms of the correspondence of the location of a physiological response to a particular anatomical structure, for example, it can be hypothesized that the blood-oxygenation-level-dependent (BOLD) fMRI response was observed in proximity of the calcarine fissure. In brain imaging techniques, analysis is often carried

out in parallel for different voxels. Then, the voxels showing a response are identified and compared with anatomical maps. The methodologies used to determine which voxels (or set of voxels) show signs of functional activation (i.e., changes in blood flow or BOLD signal) vary conspicuously. In most cases, some type of statistic is derived for each voxel (or group of voxels). For methods based on hemodynamic or metabolic phenomena, this statistic often involves comparisons across conditions. In this case, the voxels identified are those that show differences between conditions exceeding some criterion, often expressed in terms of probability of alpha error.

In most cases the response is observed over several contiguous voxels. Indeed, given the high probability of false positive in brain imaging studies, it is often the case that only responses extending over a number of voxels are considered meaningful. In this case, the exact location of the activity may be more difficult to identify. One procedure is to consider the geometrical center of the region (or group of voxels) that shows significant effects. An alternative solution is to interpret the data as indicating a response that extends over the whole area where a significant response was observed. There are two possible problems with the latter interpretation.

First, the size of the area where a significant response is observed depends on the criterion used for considering an effect as significant and on the power of the measurements. Therefore, the activated area may depend on the number of trials or number of subjects used in the study. Paradoxically, if enough trials were collected (or the signal-to-noise ratio was sufficiently high), larger areas of the brain could show signs of activation and the response would become less localized.

Second, for some measures, such as BOLD-fMRI measures, a very large activity in one voxel may actually cause other voxels (both contiguous and noncontiguous) to show activity. For this reason a strong activity localized to a very small volume in space produces effects that are difficult to distinguish from those of a weaker response localized to a larger area.

The statistical analysis of spatial information is sometimes carried on considering location as a categorical independent variable (or a factor). For instance, analysis of ERP scalp distribution is often carried out using electrode location as a factor in a factorial ANOVA design. Effects of experimental variables on scalp distribution would then be visible as interaction between these factors. Although quite popular, this approach has been criticized for two reasons.

First, the ANOVA model is an additive model, so that the contribution of electrode location as a factor to the overall variance is considered as independent from other factors. However, because ERPs always reflect the difference in voltage between locations, if a phenomenon influences the voltage of an ERP activity, it is likely to influence different electrode locations in a different manner. Therefore, an interaction between electrode location and any experimen-

tal factor may be due to a change in the overall size of a particular ERP activity, change in its spatial distribution (or scalp distribution), or interaction between these factors. McCarthy and Wood (1985) have proposed that, before submitting ERP data to an ANOVA, the values obtained at different electrodes be re-scaled by a factor related to the absolute size of the activity (e.g., the range of values or the standard deviation across all electrode locations). This "standardization" process is intended to eliminate variance across conditions or subjects due to changes in component amplitude, so that any effects observed as a function of electrode location can be interpreted as a change in the spatial distribution of the ERP activity. This approach is useful in those cases in which the ERP observed at the scalp (at all scalp locations) is determined by a single component. However, as shown in Figure 35.9, in the case of overlapping components, the standardization procedure does not eliminate the confounding between differences in scalp distribution and differences in component amplitude.

Second, the ERP values measured at different electrode locations are in fact observations of the same phenomenon from different vantage points. For this reason, both the signal and the noise can be correlated. The correlation of the noise observed at different electrode locations (or at different data points) generates a problem in the ANOVA model that is commonly known as "lack-of-sphericity" problem (Jennings & Wood, 1976). A solution to this problem is to use a multivariate approach to the analysis of scalp distribution data. Several multivariate approaches have been proposed. They include: MANOVA (Vasey & Thayer, 1987); multiple regression (Gratton, Coles et al., 1989); principal component analysis or single-value decomposition (Lamothe & Stroink, 1991); and independent component analysis (Makeig et al., 1997).

Other investigators have considered using nonparametric statistics to evaluate location information. One such example is the use of bootstrapping procedures for studying the location of the peak of surface distribution of ERPs or optical activity (Fabiani, Gratton, Friedman, & Corballis, 1998). The purpose of the bootstrap method proposed by Fabiani and colleagues is to estimate the reliability of maxima of maps of ERP or EROS activity obtained on average data. The bootstrap method involves analysis of a large number of samples of individual trials (bootstrap replications). Each bootstrap replication is obtained by extracting at random (and with replacement, so that a given trial can occur more than once) a number of trials from the original data set equal to the number of trials used to compute the regular average. For each new bootstrap replication, an average map is computed and the location of maximum is determined. The distribution of the maxima across a large number of different bootstrap replications is then obtained. The location of a maximum is reliable if it is obtained in a large proportion (e.g., 95%) of the bootstrap replications. This bootstrap method does not require assumptions about the distribution of

Effect of standardization on ERP scalp distribution data

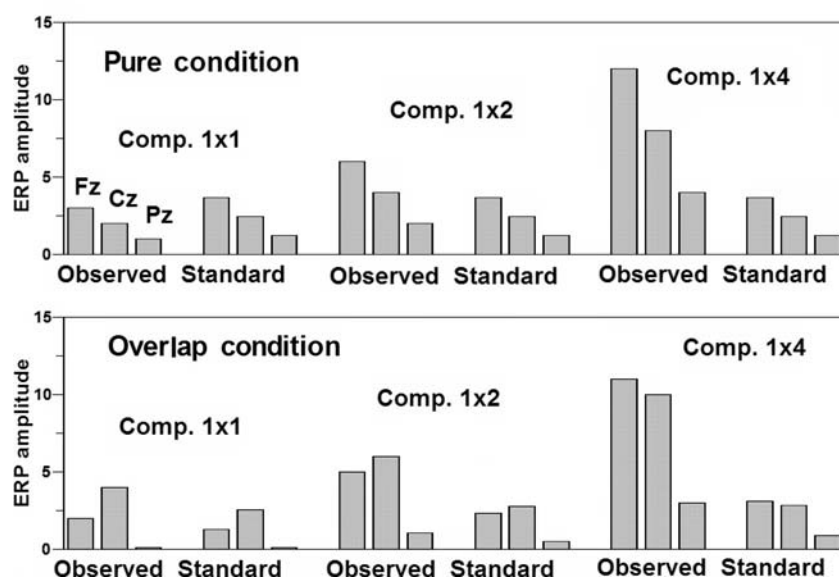


Figure 35.9. Effect of standardizing ERP scalp distribution data in conditions with different component amplitude and component overlap. **Top row:** data from a condition in which there is no component overlap (i.e., all the activity obtained at all electrode location can be attributed to one component). On the left are plots of the observed and standardized scalp distribution data from a condition in which the component is small. Standardization is obtained by dividing the data by the standard deviation across electrode locations. In the middle are similar plots for a condition in which the component has medium size. On the right are plots for a condition in which the component has large size. Note that plots of the standardized scalp distributions are identical for all component sizes. **Bottom row:** plot from conditions with component overlap. A component with central-maximum scalp distribution and fixed amplitude was added to the observed data presented on the top row. Note that the scalp distribution appears to change from a frontal to a central maximum as a function of component amplitude. Also note that the difference in scalp distribution is not eliminated by standardization.

maxima in the population, and is therefore quite robust. The method can be modified to consider the probability that the maximum is within a certain area. A limitation of the method is that it is “ad hoc,” though other bootstrap procedures can be used for different applications (Wasserman & Bockenholt, 1989). Other investigators have proposed other “distribution-free” methods for the analysis of spatial distribution data (Karniski, Blair, & Snider, 1994).

4.1.3. Intensity

Intensity is usually quantified using one of four approaches: (a) peak amplitude, (b) integrated activity over time (also called “area measure”), (c) covariance with a template (Fabiani et al., 1987), and (d) frequency of response. These measures are illustrated in Figure 35.10.

Peak and area measures are heavily dependent on the definition of a “baseline” level (i.e., these measures are taken by computing the difference between the peak, or the sum of the individual values, and the baseline; see section 4.3.1 for a discussion of baseline levels in psychophysiological measures), whereas the covariance measure requires some hypotheses about the shape of the response to be

studied. The response shape can also be obtained with statistical methods such as PCA or ICA. Area measures are less sensitive to high frequency noise than peak measures. However, empirical studies have shown that if a low-pass filter is applied before the peak measurement, the latter can be at least as reliable as area measures (Fabiani et al., 1987). Covariance measures can be used to separate the contribution of overlapping physiological responses, in which case a multiple regression approach can be used, and is quite reliable. Examples of covariance measures are given by estimates of the amplitude of different components of the ERP obtained with PCA (Donchin & Heffley, 1978) or with step-wise discriminant analysis (SWDA; Donchin, 1969; Donchin & Herning, 1975; Squires & Donchin, 1976). In these cases the covariance measures are taken with respect to a “weighting” function that has been optimized on the basis of statistical properties of the data (see the next section). Multiple regression approaches have been used to analyze hemodynamic data in complex, factorial designs (e.g., fMRI and PET). An advantage of this approach, compared to the more traditional “subtraction” methods used to analyze these data, is that it is possible to identify the effects associated with interaction terms on the brain image maps. However, the assumptions

underlying these methods – linearity and additivity – are as critical for these measures as they are for subtraction methods (see Sections 2.2 and 4.3.2).

All these measures are assumed to prove results along interval scales, which can be analyzed using parametric statistics. The error distribution tends to be skewed because of the fact that the range is usually limited at the lower end, but platykurtosis is not a problem as it was for latency measures. Amplitude measures may therefore be more reliable than latency measures (Fabiani et al., 1987).

Measures of the frequency of response are used for signals that are quite stereotyped and easily recognizable, but occur on a certain proportion of the trials or at irregular intervals. Note that, in many cases (e.g., blink or electrodermal responses), response frequency will not follow a normal distribution, and the distributions will be heavily positively skewed with some subjects exhibiting a large number of responses and others very few or none. Experimental variables will generally produce larger effects for the subjects exhibiting a greater base-rate of responses than for those exhibiting a smaller base-rate. For instance,

Measurement of the amplitude of a psychophysiological response

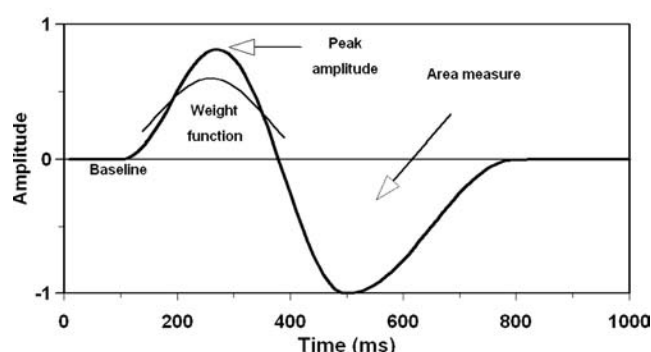


Figure 35.10. Different measures of the intensity of a psychophysiological response. Three measures are illustrated: peak amplitude, area measure, and weighted amplitude. Note that peak amplitude and area measures require a definition of the baseline value. Weighted amplitude measures can be made independent of the definition of the baseline.

all subjects may double their response frequency as a function of experimental manipulations. However, for those with a low base-rate this may generate a very small effect, whereas for those with a high base-rate the effect may be very large. This results in a large “subject by manipulation” interaction with a consequent reduced statistical power. In such cases, a logarithmic transformation of the response frequency may help increase the statistical power of the analysis. Alternatively, inferential analysis also can be conducted using a log-linear approach (Kennedy, 1984). For a discussion of other approaches, including standardization and re-scaling, see Ben-Shakhar (1985) and Stemmler (1987).

4.2. Statistical approaches to data reduction

As mentioned above, psychophysiological data are essentially multivariate. One of the major steps in the analysis is the reduction of the number of dependent variables, the number of which is often bewildering, to a smaller number appropriate for inferential analysis. Statisticians have developed a number of procedures to deal with the problem of data reduction on the basis of the statistical properties of the data. One of the simplest and most popular approaches is PCA (Donchin & Heffley, 1977). PCA is a statistical procedure for grouping dependent variables in a smaller subset of underlying (or “latent”) variables that possess the following properties: (a) they explain the variance and covariance of the original set of dependent variables as much as possible, and (b) are orthogonal to each other or, in other words, are uncorrelated. PCA components can be rotated (e.g., by using the procedure of VARIMAX) to facilitate interpretability. PCA components are interpreted in terms of their correlations (or covariances) with the original variables, which are also called “component loadings.” Donchin and collaborators wrote a

series of papers outlining the application of PCA to the analysis of ERP data. In this approach, ERP data are viewed as a data matrix in which data points are viewed as variates (i.e., dependent variables in a multivariate approach) and different subjects, conditions, and electrode locations are viewed as individual observations. A standard PCA is then run on the covariance matrix obtained from this data matrix. In most cases, a VARIMAX rotation is run on the results of the PCA. The component loadings are used to describe different components of the ERPs. These loadings are then used as sets of weights to compute component scores for each subject, condition, and location. The component scores are used as estimates of the amplitude of each component for each of the original waveforms, and submitted to inferential analysis (such as ANOVA). This approach has been used in a number of studies (McCallum & Curry, 1984; Karis, Fabiani, & Donchin, 1984; Squires, Squires, & Hillyard, 1975; Ruchkin, Sutton, & Stega, 1980), mostly because it provides separate estimates of the amplitude of individual components even when they overlap in time.

This PCA/VARIMAX/ANOVA approach has been subsequently criticized (Wood & McCarthy, 1984; Möcks, 1986; Möcks & Verleger, 1985) on the grounds that variance is misattributed to different components. Indeed, in principle, the subdivision of the variance into different components as obtained with PCA is entirely arbitrary, and there exist an infinite number of other ways of subdividing the variance on the basis of the same number of components. Therefore, it is unjustified to consider the components obtained with PCA as if they were physiological entities. Further problems with the use of PCA in the analysis of ERPs include the difficulty of incorporating latency shifts within the PCA model (Möcks, 1986) and possible correlations between components. However, PCA remains an interesting attempt at a principled way for data reduction, in which the original variance is retained as much as possible while the number of independent variables is minimized. This, as already mentioned, is one of the major goals of signal processing.

Recently, similar approaches have been proposed to address the issue of data reduction. For instance, Maier, Dagnelie, Spekreijse, and van Dijk (1987) have used a “spatial” PCA, in which the dependent variables used are spatial locations rather than data points, as a preliminary step for source analysis of ERPs. Spatial PCA has also been used to determine the number of active components at a certain moment in time, as a preliminary step for dipole analysis (see BESA; Scherg & von Cramon, 1986). A sequence of spatial and temporal PCAs have been recently employed by Spencer, Dien, and Donchin (1997), whereas Möcks (1988) has proposed a multivariate approach that also combines spatial and temporal information. Singular-value decomposition has also been proposed as an alternative to PCA (see EMSE; Greenblatt, 1996). A problem with most of these alternative approaches, just as for PCA, is that the results are not “unique” because an infinite number of

alternative descriptions of the data are possible. Furthermore, it is often difficult to interpret the results of the various data reduction procedures used and to compare results obtained in different experiments.

Independent component analysis (Makeig et al., 1997) has also become a popular method for data reduction. As with PCA, ICA is a method to decompose the variance of a set of data into a set of underlying components. It is often associated with the problem of “blind source separation,” which refers to the issue of separating simultaneous signals from a time series, for example, two speech signals occurring at the same time. This problem may be considered similar to separating different concomitant psychophysiological signals. However, in contrast to PCA, components are not selected by hierarchically maximizing the amount of variance they account for. Rather, the components are selected so as to maximize their statistical independence. The extraction procedure is more complex than in PCA. Several methods are available, which typically involve a stepwise procedure. ICA has been applied to many psychophysiological studies, with two major purposes: (a) separating signals from noise (e.g., ERP from eye-movement artifacts); (b) separating different types of brain activities from each other.

4.3. Other quantification issues

4.3.1. Absolute versus relative effects: The problem of baseline and reference

In a number of cases, psychophysiological measures are intended to reflect a change in a physiological variable from a level that exists prior to the introduction of an experimental manipulation (e.g., the presentation of a stimulus or the administration of a drug) to a level that exists after the manipulation. The level of the physiological variable before the manipulation begins is called the “rest,” or “baseline,” level. For some variables, this level is characterized by the absence of any measurable activity (e.g., there may be little EMG activity in a muscle at rest). However, most commonly, the targeted physiological systems are already active prior to the experiment, and remain active during the experiment, which is clearly the case for the heart and the brain. In these cases, the baseline level may change as a function of subject, location, and time of measurement. For this reason, most physiological measures are expressed as changes with respect to a baseline level. Determination of an appropriate baseline level is not always simple because it is not necessarily possible to induce a pure “rest” condition. In most cases, even when the experimental condition does not require a measurable overt response, the subject may be actively predicting and evaluating situations from both a cognitive and an emotional standpoint. In addition, the subject may be paying attention to internal or external stimuli that are not experimentally controlled. All of these psychological activities may influence the measurements obtained during the baseline period. If they do so in a manner that is systematically different from that observed at the time of

the experimental measurement, or in different ways in different experimental conditions, there is the possibility that systematic effects may influence the measurements. For instance, in most experiments using ERPs, measurements are taken as differences compared with a prestimulus baseline level. However, if subjects can anticipate certain properties of the stimulus during the prestimulus period, it is possible that an anticipation component of the ERP, for example, the contingent negative variation (CNV) may be elicited during this interval (Walter et al., 1964). The differential measures taken at a later time will then be affected by the presence of a CNV during the baseline period. The same may occur if the dependent variable is heart rate. In this case, expectation for an external stimulus may induce heart rate deceleration (Lacey, Kagan, Lacey, & Moss, 1963). Note, however, that if the interest is in comparing conditions in which anticipation can be expected to be the same, this is not necessarily a critical problem because a similar value is subtracted from both conditions. Because of this and other similar reasons, it is generally easier to compare activity observed in different experimental conditions than to draw conclusions from the activity observed in a single condition alone. In other words, it is much easier to consider relative than absolute effects.

Whereas the problem of the baseline or rest level is common to most physiological measures, some measures (e.g., ERPs) entail the problem of the reference. The term “reference” is used to indicate a comparison value that is valid for all points in space and is in contrast with the term “baseline,” which is used to indicate a comparison value that is valid for all points in time. In some cases, physiological measures are obtained by considering the difference in electric potential between two locations. Electric potentials do not possess absolute values, but are defined as differences between two energy levels. Therefore, measures of electric potentials should be considered with respect to the locations that are involved. It is, however, possible to compare the difference in potential that exists between point A and point B and between point C and point B. If the first difference is more positive than the second, it can be concluded that point A has a more positive voltage level than point C. In this case point B will be used as a common reference point. If measures were taken only at a single point in time, then the point that is used as a common reference makes little difference. However, voltage measures often are taken at different times, and points of maximum difference (or peak values) are identified. The point at which A is most different from B may not be the same point at which A is most different from C. Therefore, the time of the occurrence of the peak on A may differ depending on whether B or C is used as a reference. For this reason, appropriate selection of a reference location is a critical aspect of recording electrical activity.

Historically, there have been three approaches to the problem of reference selection. The first approach is to consider measurements obtained between selected

pairs of electrode locations; the measurement is labeled “bipolar.” Bipolar measurements are common practice in the study of EMG, EKG, and EOG. In some cases, bipolar derivations are used to study local phenomena, as in the cases of EMG and EOG. In these cases, the two electrodes are placed very close to each other, so that activity generated far away is likely to influence the electrodes equally, (i.e. to cancel out each other), whereas activity generated locally is more likely to generate differences. This approach is adequate when the interest is in studying the time course of activity of relatively simple electric field configuration, such as for EMG and EOG, and, to some extent, EKG. For EEG studies, the bipolar approach fails to account for the complexities of the fields generated by the brain, and is only used in some clinical applications or special cases, for example, the recording of auditory brainstem averaged evoked potentials (BAEPs).

A second approach is to consider measurements from a number of locations as differences with respect to the same location (or to a common value). This approach is called a “common reference system.” There are a number of choices for the location of the reference value, including the ears, mastoids, nosetip, forehead, or locations outside the head (i.e., extracephalic reference). The latter may pick up electrical activity from the heart (EKG) and, therefore, require special compensation procedures (Fortgens & de Bruin, 1983). Finally, some researchers have advocated the use of an average reference, which is obtained by subtracting or adding a value to all of the observed locations, so that the algebraic sum of all of the potential values observed at different locations is equal to zero. Although arguments in favor of one or another of these systems have been proposed (Skrandies & Lehmann, 1982), there appears to be no clear advantage in using one reference system over another – the choice is arbitrary. However, because the selection of the reference may alter the shape of waveforms, including both the amplitude and the latency of the peaks, it is very important that the selection of the reference is made explicit and the same reference system is used when comparing across different data sets. The common reference system is the one most used in ERP research. Its greatest advantages are ease of computation and possibility of comparing values obtained at different locations. Its major drawback is the arbitrariness of the choice of the reference.

A third approach is that of using a transformation of the data, so that the data are expressed in terms of absolute dimensions (e.g., current flow) instead of relative dimensions (e.g., difference in potential). As seen above (Section 3.3.2), this can be achieved by computing the local gradient in potential instead of the difference with a common reference, but this requires the assumption of constant resistance between electrode locations.

4.3.2. Subtraction, comparisons, and linearity of the measures

In most cases, psychophysicists are interested in comparing physiological responses in two (or more) condi-

tions. The comparison process usually involves a study of the statistical reliability of differences between measures. This, in turn, involves computing several differences across subjects, time points, and locations. Data obtained in different conditions can be interpreted either along ordinal scales, which require using less powerful nonparametric statistics, or along interval scales, which may be analyzed using more powerful parametric statistics. The basic assumption of interval scales is that differences between intervals at any level of the scale are directly comparable. For example, the difference between the values 2 and 4 in some scale has the same significance as the difference between the values 22 and 24. This allows the investigator to subtract common terms and to compare data directly. We have previously considered a similar property called “linearity” (see Section 2.2).

Most psychophysiological measures, however, depart from linearity to some extent. One of the reasons is that the measures often have a limited range because physiological systems have feedback mechanisms that counteract extreme values in order to maintain homeostasis. For example, heart rate is very unlikely to drop below 40 beats per minute even under strong vagal activation. There are also other limitations because of the number of units (e.g., neurons) that can respond to particular stimuli. However, for most psychophysiological measures, there exists an interval of values for which linearity is valid. For practical purposes, it is advantageous that, if at all possible, measurements be taken in conditions in which the psychophysiological measures are within the range in which they exhibit linearity.

Range and scale differences may also vary as a function of subject or location on the body. For example, certain subjects may exhibit higher variability in a psychophysiological variable than others. Similarly, measures of a psychophysiological variable may have a greater range at a location than another. This will make it more difficult to compare data across subjects or locations. To counteract this problem, standardized measures in which the data are transformed, so to have equal means and standard deviations, are often used in these types of comparisons.

For a number of psychophysiological variables, it has been found that variability is correlated with the average level of the variable itself for a particular subject or condition. This relationship some time takes the form of “ceiling” or “floor” effects, which in psychophysiology often have been considered as examples of the “law of initial value” (Wilder, 1967). This would produce departure from linearity. Psychologists have often used special transformations (such as “logit” or “probit” transformations) to correct these forms of departures from linearity. In other cases, effects are proportional to the original value. In these cases, a logarithmic transformation of the data (or the division by the average value) may equalize variances (and differences) across conditions, and produce a scale that has linear properties. For a discussion of several procedures for comparing conditions with different base rates see Wainer (1991).

5. CONCLUDING REMARKS

This chapter has reviewed several issues related to the analysis of psychophysiological data. In most cases, procedures used in data analysis can be viewed as an effort to extract meaningful information from the data that are often under conditions in which the data are noisy. In these cases, the majority of the data-analyst efforts are devoted to increasing the signal-to-noise ratio. Although there are procedures that increase this ratio under all conditions, it is often the case that a priori hypotheses about what types of signals to expect will help in the design of appropriate analysis procedures. These a priori hypotheses do not necessarily have to be very specific (i.e., activity at a particular latency and at a particular location is determined by factor A or factor B). Even simple hypotheses, for example, that some activity should be observed somewhere during a particular latency interval or at a particular location, will help design appropriate analytical methods. However, in general, the power of the analytical procedure increases with the specificity of the hypotheses that are entertained.

It is often the case that psychophysiological phenomena are quite complex and unpredicted effects are obtained in a number of studies. In some cases, discovery of these new effects is what affords the greatest scientific advancements. This generates a dilemma for the data analyst – how can maximum power be obtained, while still maintaining a wide focus in the analysis? In most cases the answer is in alternating studies conducted with an “exploratory” attitude, which enables the investigators to detect unexpected findings, with other more “hypothesis-driven” studies, which provide more rigorous tests of specific hypotheses. This problem emphasizes that data analysis needs to be integrated with experimental design in a bidirectional fashion.

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